

QUESTIONS

These released questions are from the mathematics component of the junior-division assessment. This section provides the overall expectation and the category of knowledge and skills for each question.

B. NUMBER

B1. Number Sense

demonstrate an understanding of numbers and make connections to the way numbers are used in everyday life

- 1** Which number represents ninety-two thousand five hundred four?
KU

A 92 054

B 92 504

C 920 504

D 920 054

- 2** Which fraction is equivalent to $\frac{6}{5}$?
AP

A $\frac{5}{4}$

B $\frac{5}{6}$

C $1\frac{1}{5}$

D $1\frac{5}{6}$

B2. Operations

use knowledge of numbers and operations to solve mathematical problems encountered in everyday life

3 What is $5.2 \div 4$?

KU

A 0.77

B 1.3

C 9.2

D 13

4 The cost of one package of seeds is \$0.50.

AP

What is the cost of 100 packages of seeds?

A \$5

B \$50

C \$500

D \$5000

B2. Operations (continued)

5 A field is planted with wheat, oats and corn.

TH

- $\frac{5}{9}$ of the field is planted with wheat
- $\frac{7}{36}$ of the field is planted with oats

Which fraction of the field is planted with corn?

A $\frac{27}{36}$

B $\frac{12}{45}$

C $\frac{2}{27}$

D $\frac{1}{4}$

C. ALGEBRA

C1. Patterns and Relationships

identify, describe, extend, create, and make predictions about a variety of patterns, including those found in real-life contexts

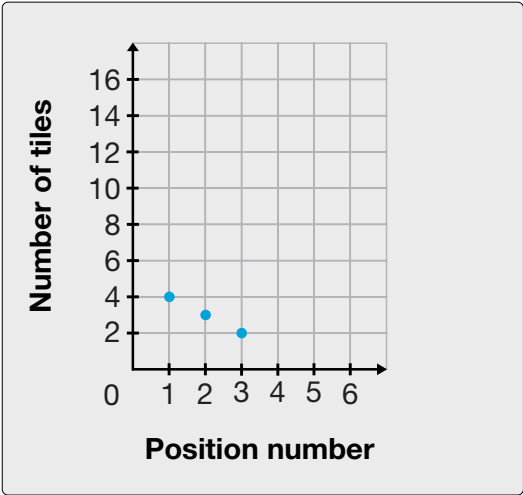
6 A pattern is made using tiles. Information about the pattern is shown in this table of values.

KU

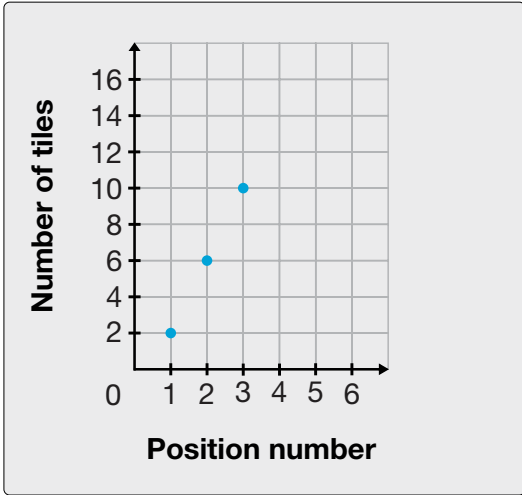
Position number	Number of tiles
1	4
2	8
3	12

Which graph represents the data in this table of values?

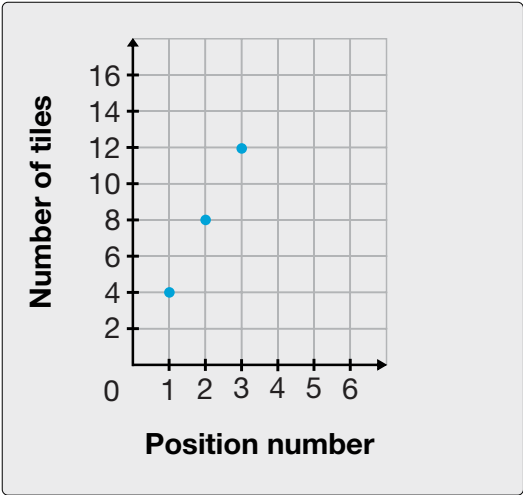
A



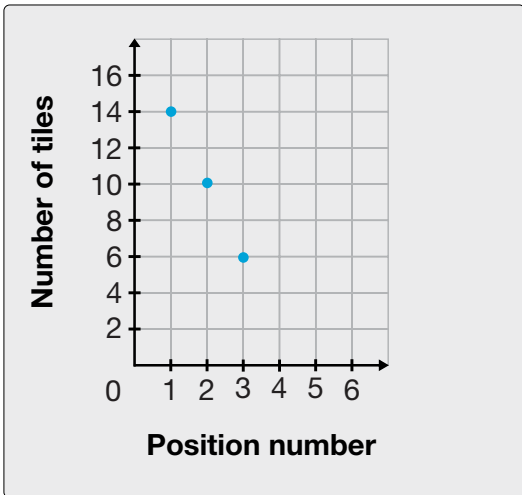
B



C



D



C1. Patterns and Relationships (continued)

7 This table of values shows a growing pattern.
AP

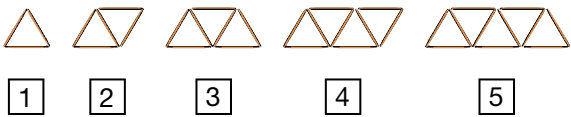
Term number	Term value
1	3
2	7
3	11
4	15

The pattern continues to increase by the same amount each time.

What will the term number be when the term value is 23?

- A 7
- B 6
- C 5
- D 4

8 Toothpicks are used to make this growing pattern.
TH



How many toothpicks are needed to make the figure in position number 25?

- A 13 toothpicks
- B 50 toothpicks
- C 51 toothpicks
- D 75 toothpicks

C2. Equations and Inequalities

demonstrate an understanding of variables, expressions, equalities, and inequalities, and apply this understanding in various contexts

9 What value of k makes this equation true?

KU

$$k - 9 = 4$$

A

2

B

5

C

13

D

36

10 Which statement is true when $x = 4$?

AP

A

$$5x - 8 = 24$$

B

$$5x - 8 > 24$$

C

$$5x - 8 < 24$$

D. DATA

D1. Data Literacy

manage, analyse, and use data to make convincing arguments and informed decisions, in various contexts drawn from real life

12 A teacher orders new magazines each year.
KU He surveys a sample of students in the school to find out their preferences.

Which of these samples would **best** represent all the students in the school for this survey?

- A 10 students from each class
- B 10 students from one class
- C 10 students in Grade 5
- D 10 students on the soccer team

13 This table represents the number of pages read by students during one week.
AP

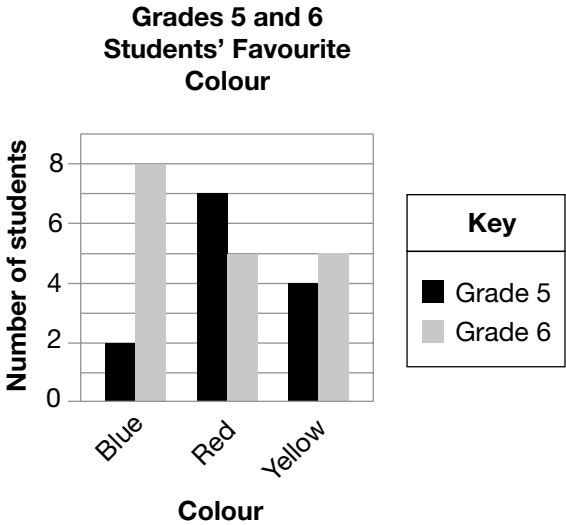
Name	Number of pages read
Zachary	75
Jacob	82
Iman	98
Florence	82
Shannon	63
Anthony	80

What is the **median** number of pages read?

- A 90 pages
- B 82 pages
- C 81 pages
- D 80 pages

D1. Data Literacy (continued)

14 Which statement about this graph is true?
TH



- A** There were 16 students in Grade 6 who answered the survey.
- B** The colour chosen the most by students who answered the survey was blue.
- C** The colour chosen the least by students who answered the survey was yellow.
- D** There were more students in Grade 5 than in Grade 6 who answered the survey.

D2. Probability

describe the likelihood that events will happen, and use that information to make predictions

15 **KU** Roger conducts an experiment where he flips a coin ten times and records whether it lands on heads or tails in a table.

He finds that his experimental probability matches the theoretical probability.

Select the table that shows the results from Roger's experiment.

A

Heads	Tails
	

B

Heads	Tails
 	

C

Heads	Tails
	

D

Heads	Tails
 	

16 **AP** A bag has 8 red tiles and 2 purple tiles.

Remi selects a tile from the bag without looking.

What is the probability that Remi selects a red tile from the bag?

- A

0.02
- B

0.08
- C

0.2
- D

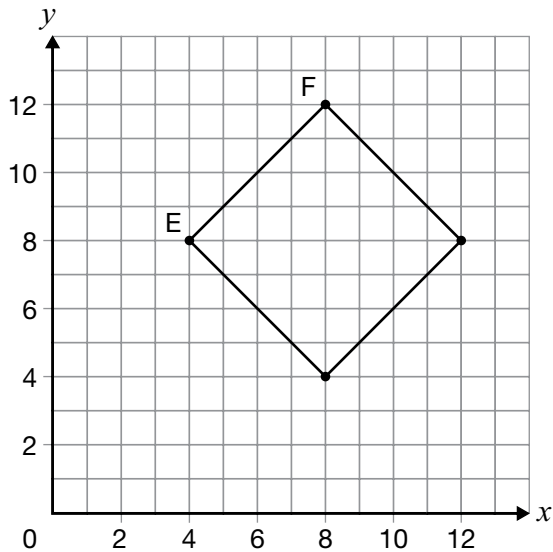
0.8

E. SPATIAL SENSE

E1. Geometric and Spatial Reasoning

describe and represent shape, location, and movement by applying geometric properties and spatial relationships in order to navigate the world around them

17 What are the coordinates of vertices E and F?
KU



- A E (4, 8) and F (8, 12)
- B E (4, 8) and F (12, 8)
- C E (8, 4) and F (8, 12)
- D E (8, 4) and F (12, 8)

18 Which quadrilateral has rotational symmetry of order 2?
AP

- A a kite
- B a square
- C a dart
- D a rectangle

E2. Measurement

compare, estimate, and determine measurements in various contexts

19 How many grams are equal to 7.2 kg?

KU

A 72 000 g

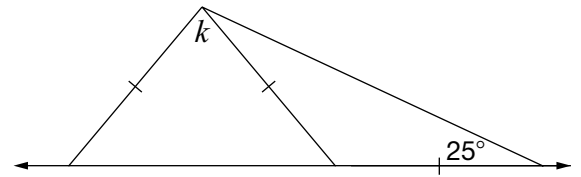
B 7200 g

C 720 g

D 72 g

20 What is the value of k ?

TH



A 50°

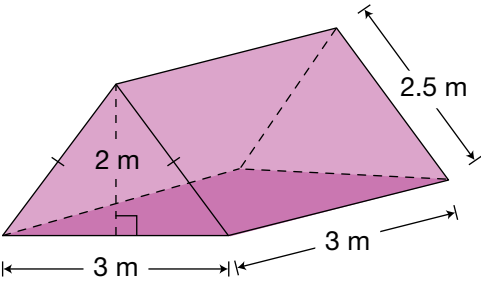
B 65°

C 80°

D 130°

E2. Measurement (continued)

21 What is the surface area of this triangle-based prism?
TH



- A 10.5 m²
- B 21 m²
- C 30 m²
- D 36 m²